GBU 2301: Business Statistics

Week 6: Discrete Probability Distributions II

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Required reading: Anderson et al., Statistics for Business and Economics, 15th ed., Sections 5.5–5.7

Outline

- Binomial Probability Distribution
- Poisson Probability Distribution
- Hypergeometric Probability Distribution

Binomial Probability Distribution

- Two customers, each with a purchase probability of 0.40
- Independent purchase decisions
- X: number of purchases

x	Outcomes	$P(X = x)$
0	No purchase, no purchase	$0.60^2 = 0.36$
1	Purchase, no purchase; no purchase, purchase	$2(0.40)(0.60) = 0.48$
2	Purchase, purchase	$0.40^2 = 0.16$

- Two customers: manageable to list every outcome
- Ten customers: $2^{10} = 1{,}024$ purchase/no-purchase sequences
- Binomial model: probabilities for the purchase count without listing every sequence

Binomial Probability Distribution

Let X count the number of successes in n trials. A binomial model requires:

Condition	Meaning
Fixed number of trials	The number n is set before observing the outcomes.
Two outcomes per trial	Each trial is classified as success or failure.
Constant probability	The success probability p is the same on every trial.
Independent trials	One trial's outcome does not change the probabilities for the others.

$$X \sim \text{Bin}(n, p), \qquad x = 0, 1, \ldots, n.$$

- “Success”: the outcome being counted
  - If X counts defective items, a defect is a success in this notation.
- For example, each customer has a 40% chance of buying.
  - Knowing whether one customer buys does not change the 40% probability for another customer.

Binomial Probabilities: One Sequence

- Three customers
- Independent decisions and a common purchase probability

Number of customers, n	Purchase probability, p	No-purchase probability, $1-p$
3	0.40	0.60

Let B denote a purchase and N no purchase. The probability of the particular sequence (B, B, N) is

$$P(B, B, N) = 0.40(0.40)(0.60) = 0.096.$$

Exactly two purchases can occur in three ways:

Sequence	Probability
(B, B, N)	$0.40(0.40)(0.60) = 0.096$
(B, N, B)	$0.40(0.60)(0.40) = 0.096$
(N, B, B)	$0.60(0.40)(0.40) = 0.096$

$$P(X = 2) = 0.096 + 0.096 + 0.096 = 0.288.$$

The probability of one sequence is not the probability of the count.

Binomial Coefficient

- x successes in n trials: choose the x success positions.
- Remaining $n - x$ positions: failures

$$\text{Number of sequences} = \binom{n}{x} = \frac{n!}{x!(n-x)!}, \qquad 0! = 1.$$

For two purchases among three customers:

$$\binom{3}{2} = \frac{3!}{2!1!} = \frac{6}{2} = 3.$$

Each of these sequences has probability

$$p^x(1-p)^{n-x}.$$

The sequences are mutually exclusive, so their probabilities add:

$$P(X = x) = \underbrace{\binom{n}{x}}_{\text{number of sequences}} \underbrace{p^x(1-p)^{n-x}}_{\text{probability of each sequence}}.$$

For no successes or all successes, there is only one sequence: $\binom{n}{0} = \binom{n}{n} = 1$.

Binomial Probability Function

For $X \sim \text{Bin}(n, p)$,

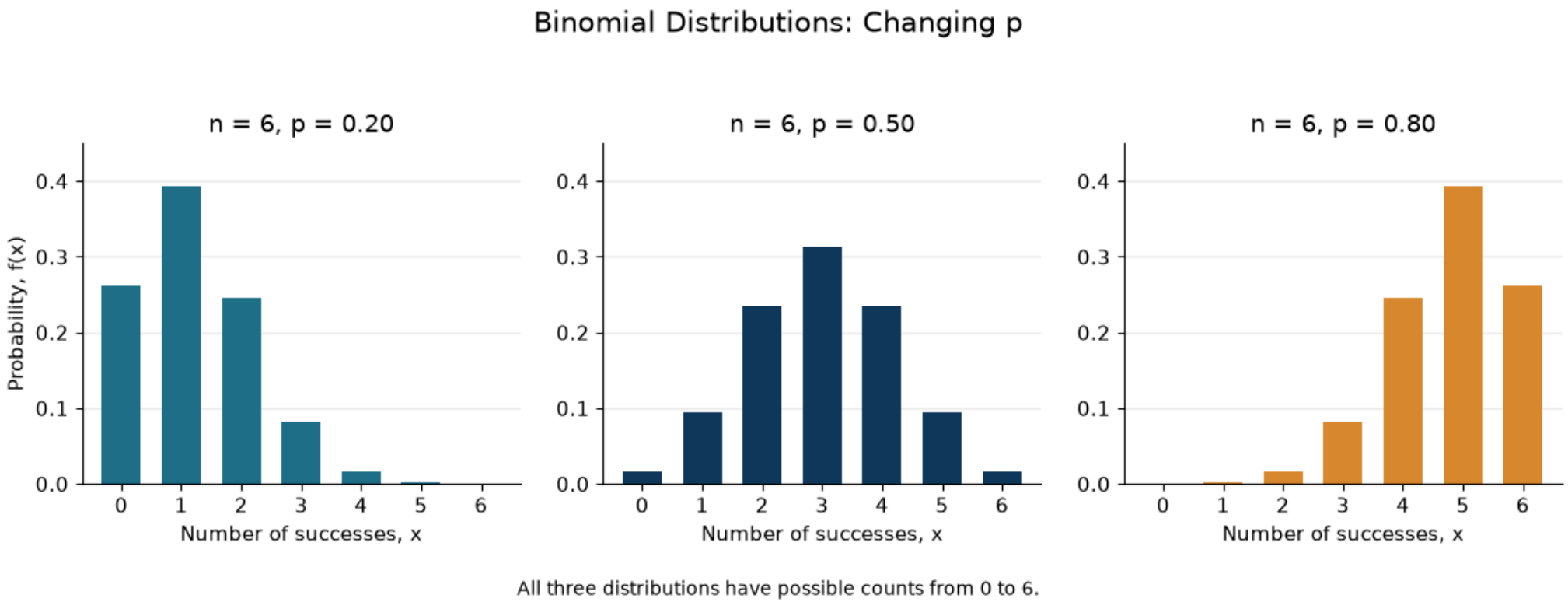
$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

For the three customers with $p = 0.40$:

Purchases, x	Number of sequences	Probability of each sequence	$f(x)$
0	$\binom{3}{0} = 1$	$0.40^0(0.60)^3 = 0.216$	0.216
1	$\binom{3}{1} = 3$	$0.40^1(0.60)^2 = 0.144$	0.432
2	$\binom{3}{2} = 3$	$0.40^2(0.60)^1 = 0.096$	0.288
3	$\binom{3}{3} = 1$	$0.40^3(0.60)^0 = 0.064$	0.064
Total	8		1.000

- Next graphs: fixed $n = 6$, different values of p
- Same possible counts; different probabilities

Binomial Probability Function



All three distributions have possible counts from 0 to 6.

Binomial Probabilities: Sales

A broker contacts five potential customers.

- Independent decisions
- Same sale probability for every contact

Contacts, n	Sale probability, p	Random variable, X
5	0.40	Number of sales

The probability of exactly three sales is

$$P(X = 3) = \binom{5}{3}(0.40)^3(0.60)^2 = \frac{5!}{3!2!}(0.064)(0.36) = 10(0.064)(0.36) = 0.2304.$$

Applying the same formula to every possible count gives:

Sales, x	0	1	2	3	4	5
$f(x)$	0.07776	0.25920	0.34560	0.23040	0.07680	0.01024

$$\sum_{x=0}^{5} f(x) = 1.$$

The individual sale probability 0.40 is not the probability of exactly three sales.

Cumulative and Complementary Probabilities

For the five contacts with $p = 0.40$:

Event	Calculation	Probability
At most one sale	$P(X \le 1) = f(0) + f(1)$	$0.07776 + 0.25920 = 0.33696$
At least two sales	$P(X \ge 2) = 1 - P(X \le 1)$	$1 - 0.33696 = 0.66304$
More than two sales	$P(X > 2) = 1 - P(X \le 2)$	$1 - 0.68256 = 0.31744$
Between two and four sales, inclusive	$P(2 \le X \le 4) = 1 - f(0) - f(1) - f(5)$	0.65280

For an integer count and integer k,

$$P(X \ge k) = 1 - P(X \le k - 1), \qquad P(X > k) = 1 - P(X \le k).$$

For an inclusive interval,

$$P(a \le X \le b) = \sum_{x=a}^{b} f(x).$$

Check which boundary values belong to the event before adding or subtracting probabilities.

Cumulative and Complementary Probabilities

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Binomial Probabilities: Sales

$X \sim \text{Bin}(5, 0.40)$: at most 1 sale.

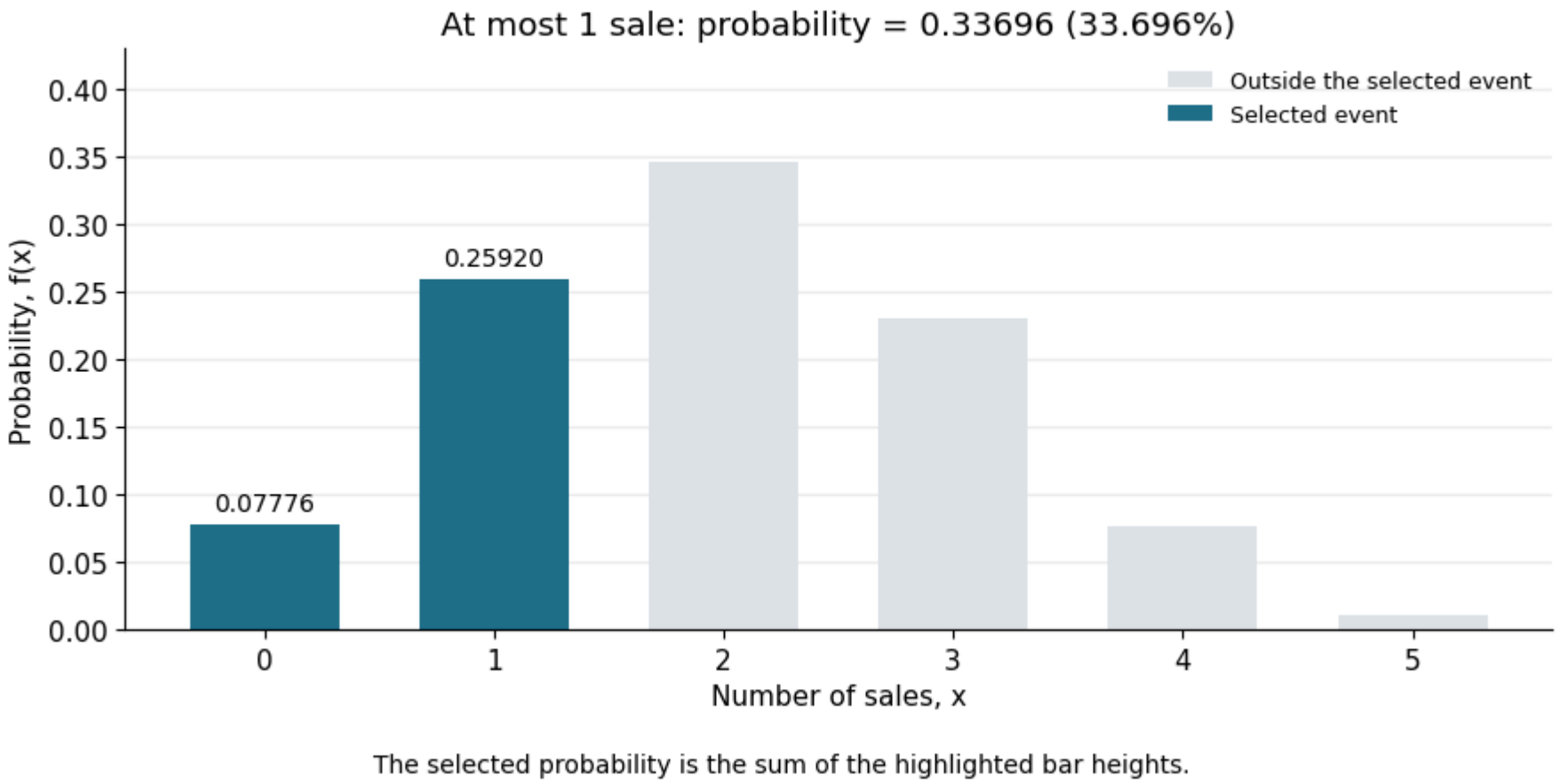
$$f(x) = \binom{5}{x}(0.40)^x(0.60)^{5-x}.$$

$$P(X \leq 1) = f(0) + f(1).$$

Sales, x	Calculation of $f(x)$	Probability
0	$\binom{5}{0}(0.40)^0(0.60)^5 = 1(0.40)^0(0.60)^5$	0.07776
1	$\binom{5}{1}(0.40)^1(0.60)^4 = 5(0.40)^1(0.60)^4$	0.25920

$$P(X \leq 1) = 0.07776 + 0.25920 = 0.33696 = 33.696\%.$$

Cumulative and Complementary Probabilities



The selected probability is the sum of the highlighted bar heights.

Exercise 1

Delta Deliveries handles four parcels. Assume that on-time deliveries are independent and each parcel has the same probability of arriving on time.

Parcels	Probability of on-time delivery per parcel
4	0.70

a. Find the probability that exactly three parcels arrive on time.

b. Find the probability that at least three parcels arrive on time.

c. Find the probability that at most one parcel arrives on time.

Exercise 1: Solution

Let X count parcels arriving on time. Then $X \sim \text{Bin}(4, 0.70)$.

$$f(x) = \binom{4}{x}(0.70)^x(0.30)^{4-x}.$$

a. Exactly three:

$$P(X = 3) = \binom{4}{3}(0.70)^3(0.30) = 4(0.343)(0.30) = 0.4116.$$

b. At least three includes three or four:

$$P(X = 4) = \binom{4}{4}(0.70)^4(0.30)^0 = 0.2401.$$

$$P(X \geq 3) = P(X = 3) + P(X = 4) = 0.4116 + 0.2401 = 0.6517.$$

c. At most one includes zero or one:

$$P(X = 0) = (0.30)^4 = 0.0081,$$

$$P(X = 1) = \binom{4}{1}(0.70)(0.30)^3 = 4(0.70)(0.027) = 0.0756.$$

$$P(X \leq 1) = 0.0081 + 0.0756 = 0.0837.$$

Exercise 2

Gamma Components tests the next six items from a production line. Assume that defect statuses are independent and the defect probability stays constant.

Items tested	Defect probability per item
6	0.10

A worksheet reports a probability of 0.006561 that exactly two of the six items are defective.

a. Check the worksheet entry. Explain what calculation could have produced it and give the correct probability.

b. The six-item batch is sent for rework if at least two items are defective. Find the probability of rework.

Exercise 2: Solution

Let X be the number of defective items. Then $X \sim \text{Bin}(6, 0.10)$.

The reported number is

$$0.10^2(0.90)^4 = 0.006561.$$

- Reported value: two specified positions defective, the other four non-defective
- Possible pairs of defective positions: $\binom{6}{2} = 15$

$$P(X = 2) = \binom{6}{2}(0.10)^2(0.90)^4 = 15(0.006561) = 0.098415.$$

For rework, include every count from two through six:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1).$$

$$P(X = 0) = 0.90^6 = 0.531441,$$

$$P(X = 1) = 6(0.10)(0.90)^5 = 0.354294.$$

$$P(\text{rework}) = 1 - 0.531441 - 0.354294 = 0.114265.$$

The rework probability is about 11.43%, not the 9.84% probability of exactly two defects.

Binomial Mean and Variance

For trial i, let $I_i = 1$ for success and $I_i = 0$ for failure. The total number of successes is

$$X = I_1 + I_2 + \cdots + I_n.$$

For one trial, $P(I_i = 1) = p$ and $P(I_i = 0) = 1 - p$.

$$E(I_i) = 0(1-p) + 1(p) = p.$$

$$\begin{aligned} \mathrm{Var}(I_i) &= (0-p)^2(1-p) + (1-p)^2 p \\ &= p^2(1-p) + p(1-p)^2 \\ &= p(1-p)\big[p + (1-p)\big] = p(1-p). \end{aligned}$$

Thus,

$$E(X) = \sum_{i=1}^{n} E(I_i) = \underbrace{p + \cdots + p}_{n \text{ terms}} = np.$$

For the variance, independence makes every covariance between different trials zero:

$$\begin{aligned} \mathrm{Var}(X) &= \sum_{i=1}^{n} \mathrm{Var}(I_i) + 2\sum_{i<j} \mathrm{Cov}(I_i, I_j) \\ &= np(1-p) + 0. \end{aligned}$$

$$\mu = np, \qquad \sigma^2 = np(1-p), \qquad \sigma = \sqrt{np(1-p)}.$$

Binomial Mean and Variance

For the broker's five contacts:

$$\mu = 5(0.40) = 2 \text{ sales}, \quad \sigma^2 = 5(0.40)(0.60) = 1.20 \text{ sales}^2, \quad \sigma = \sqrt{1.20} \approx 1.095 \text{ sales}.$$

Bernoulli Probability Distribution

- Binomial: number of successes across n trials
- Bernoulli: outcome of a single success/failure trial

Record that single trial using I:

Outcome	I	Probability
Failure	0	$1-p$
Success	1	p

A Bernoulli random variable is a binomial random variable with $n = 1$:

$$I \sim \text{Bin}(1, p).$$

The variables $I_1, \ldots, I_n$ in the preceding binomial calculation are Bernoulli variables. Their mean and variance explain the terms p and $p(1-p)$ used there.

Bernoulli Mean and Variance

Represent one trial by I, where $I = 1$ for success and $I = 0$ for failure. This is a Bernoulli random variable.

i	$P(I = i)$	$iP(I = i)$	$i^2 P(I = i)$
0	$1 - p$	0	0
1	p	p	p

$$E(I) = 0(1 - p) + 1(p) = p.$$

Since $I^2 = I$ for both possible outcomes,

$$E(I^2) = 0^2(1 - p) + 1^2 p = p.$$

Using the alternative variance formula,

$$\text{Var}(I) = E(I^2) - [E(I)]^2 = p - p^2 = p(1 - p).$$

Bernoulli Distribution: Examples

Single trial	$I = 1$	$I = 0$	Success probability, p
One customer	Buys	Does not buy	0.40
One parcel	Arrives on time	Arrives late	0.70

For the customer:

$$E(I) = 0(0.60) + 1(0.40) = 0.40,$$
$$\text{Var}(I) = 0.40(1 - 0.40) = 0.24.$$

For the parcel:

$$E(I) = 0(0.30) + 1(0.70) = 0.70,$$
$$\text{Var}(I) = 0.70(1 - 0.70) = 0.21.$$

Each trial produces either 0 or 1. The mean describes the long-run average across repeated trials.

Sales and Net Profit

Retain the broker's sales count $X \sim \text{Bin}(5, 0.40)$.

Item	Amount
Net revenue per sale, after sale-related costs	80 MAD
Fixed cost of contacting the five customers	120 MAD

Net profit is $Y = 80X - 120$. Using $E(X) = 2$ and $\text{Var}(X) = 1.20$:

$$E(Y) = 80E(X) - 120 = 80(2) - 120 = 40 \text{ MAD}.$$

$$\text{Var}(Y) = 80^2(1.20) = 7{,}680 \text{ MAD}^2,$$

$$\sigma_Y = 80\sqrt{1.20} \approx 87.64 \text{ MAD}.$$

A loss occurs when

$$80X - 120 < 0 \quad \Rightarrow \quad X < 1.5.$$

Because X is an integer,

$$P(Y < 0) = P(X \leq 1) = 0.33696.$$

Positive expected profit does not rule out a substantial probability of loss.

Exercise 3

A sales team can choose one of two campaigns.

- Within each campaign: independent decisions and a constant sale probability
- All campaign costs included below

Item	Campaign Beta	Campaign Gamma
Number of contacts	4	8
Sale probability per contact	0.50	0.25
Net revenue per sale	80 MAD	80 MAD
Fixed campaign cost	120 MAD	120 MAD

a. Calculate expected net profit and its standard deviation for each campaign.

b. A campaign is acceptable only if expected net profit is at least 30 MAD and the probability of a nonnegative net profit is at least 65%. Which campaign, if either, meets both requirements? Support the decision with calculations.

Exercise 3: Solution

For the sales counts:

$$E(X_B) = 4(0.50) = 2, \qquad E(X_G) = 8(0.25) = 2.$$

$$\text{Var}(X_B) = 4(0.50)(0.50) = 1, \qquad \text{Var}(X_G) = 8(0.25)(0.75) = 1.50.$$

Both profits have the form $Y = 80X - 120$.

Measure	Beta	Gamma
Expected net profit (MAD)	$80(2) - 120 = 40$	$80(2) - 120 = 40$
Profit SD (MAD)	$80\sqrt{1} = 80$	$80\sqrt{1.50} \approx 97.98$

Nonnegative profit requires $80X - 120 \geq 0$, so $X \geq 2$.

$$P(Y_B \geq 0) = 1 - (0.50)^4 - 4(0.50)(0.50)^3 = 0.6875.$$

$$P(Y_G \geq 0) = 1 - (0.75)^8 - 8(0.25)(0.75)^7 \approx 0.6329.$$

- Beta: both requirements met, $40 \geq 30$ and $0.6875 \geq 0.65$
- Gamma: expected-profit requirement met; probability requirement not met
- Equal expected profits do not imply equal risk.

Poisson Probability Distribution

A Poisson random variable counts occurrences over a specified interval of time or space.

Situation	Random variable
Calls reaching a support desk	Number of calls in 30 minutes
Flaws in a cable	Number of flaws in 250 metres
Printing marks on a sheet	Number of marks in one square metre

Model assumptions:

- Constant average rate over the interval
- Independent counts in non-overlapping intervals
- Individual occurrences
  - In a sufficiently short interval, two or more arrivals are negligible relative to one.

The possible counts are $0, 1, 2, \ldots$, with no fixed upper bound.

A count is not automatically Poisson. Potential violations:

- Scheduled arrivals
- Groups arriving together
- Substantial changes in the arrival rate

Poisson Probability Function

Let μ be the expected number of occurrences in the interval being studied.

$$X \sim \text{Poisson}(\mu), \qquad f(x) = P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \quad x = 0, 1, 2, \ldots.$$

- $e \approx 2.71828$ and $0! = 1$
- Use the calculator's e^z function, not a rounded value of e.
- λ: constant occurrence rate per unit of time, length, or area
- t: interval size, in units matching λ

$$\mu = \lambda t.$$

Beta Support receives calls according to a Poisson process:

Rate	Interval of interest	Expected count, μ
6 calls per hour	30 minutes $= 0.5$ hour	$6(0.5) = 3$

Therefore, the number of calls in 30 minutes has distribution $\text{Poisson}(3)$, not $\text{Poisson}(6)$.

Poisson Probabilities: Calls

For Beta Support's 30-minute interval, $\mu = 3$.

- Exactly four calls:

$$P(X = 4) = \frac{3^4 e^{-3}}{4!} = \frac{81e^{-3}}{24} \approx 0.168031.$$

- No calls:

$$P(X = 0) = \frac{3^0 e^{-3}}{0!} = e^{-3} \approx 0.049787.$$

- At least one call:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-3} \approx 0.950213.$$

- At most one call:

$$P(X \leq 1) = P(X = 0) + P(X = 1) = e^{-3} + 3e^{-3} = 4e^{-3} \approx 0.199148.$$

An average of three calls does not mean that three is the largest possible count.

Poisson Probabilities: Time Intervals

Keep the arrival rate at six calls per hour. For each interval, let X count calls during that interval.

Interval	Length in hours	μ	Probability of no calls
10 minutes	$1/6$	$6(1/6) = 1$	$e^{-1} \approx 0.367879$
30 minutes	$1/2$	$6(1/2) = 3$	$e^{-3} \approx 0.049787$
60 minutes	1	$6(1) = 6$	$e^{-6} \approx 0.002479$

- Expected count: proportional to interval length
- Probabilities: not proportional to interval length

For example,

$$P(\text{no calls in 60 minutes}) = e^{-6} = (e^{-3})^2,$$

- No calls in 60 minutes requires no calls in both 30-minute intervals.
- Non-overlapping intervals: independent counts

Doubling the interval does not double the probability of no calls.

Poisson Probabilities: Length

Gamma Cables models cable flaws by a Poisson process.

- Constant flaw rate
- Independent counts on non-overlapping lengths

Average flaw rate	Length inspected
0.8 flaws per 100 metres	250 metres

For X, the number of flaws in the inspected length,

$$\mu = \frac{0.8 \text{ flaws}}{100 \text{ metres}}(250 \text{ metres}) = 2.$$

The probability of no flaws is

$$P(X = 0) = e^{-2} \approx 0.135335.$$

The probability of at least two flaws is

$$\begin{aligned} P(X \geq 2) &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - e^{-2} - 2e^{-2} \\ &= 1 - 3e^{-2} \approx 0.593994. \end{aligned}$$

The exposure is cable length rather than time; the same probability function applies.

Poisson Probabilities: Length

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Poisson Probabilities: Calls

Beta Support: 6 calls per hour. At most 1 call in 30 minutes.

$$\mu = 6\left(\frac{30}{60}\right) = 3, \qquad X \sim \text{Poisson}(3).$$

$$f(x) = \frac{3^x e^{-3}}{x!}.$$

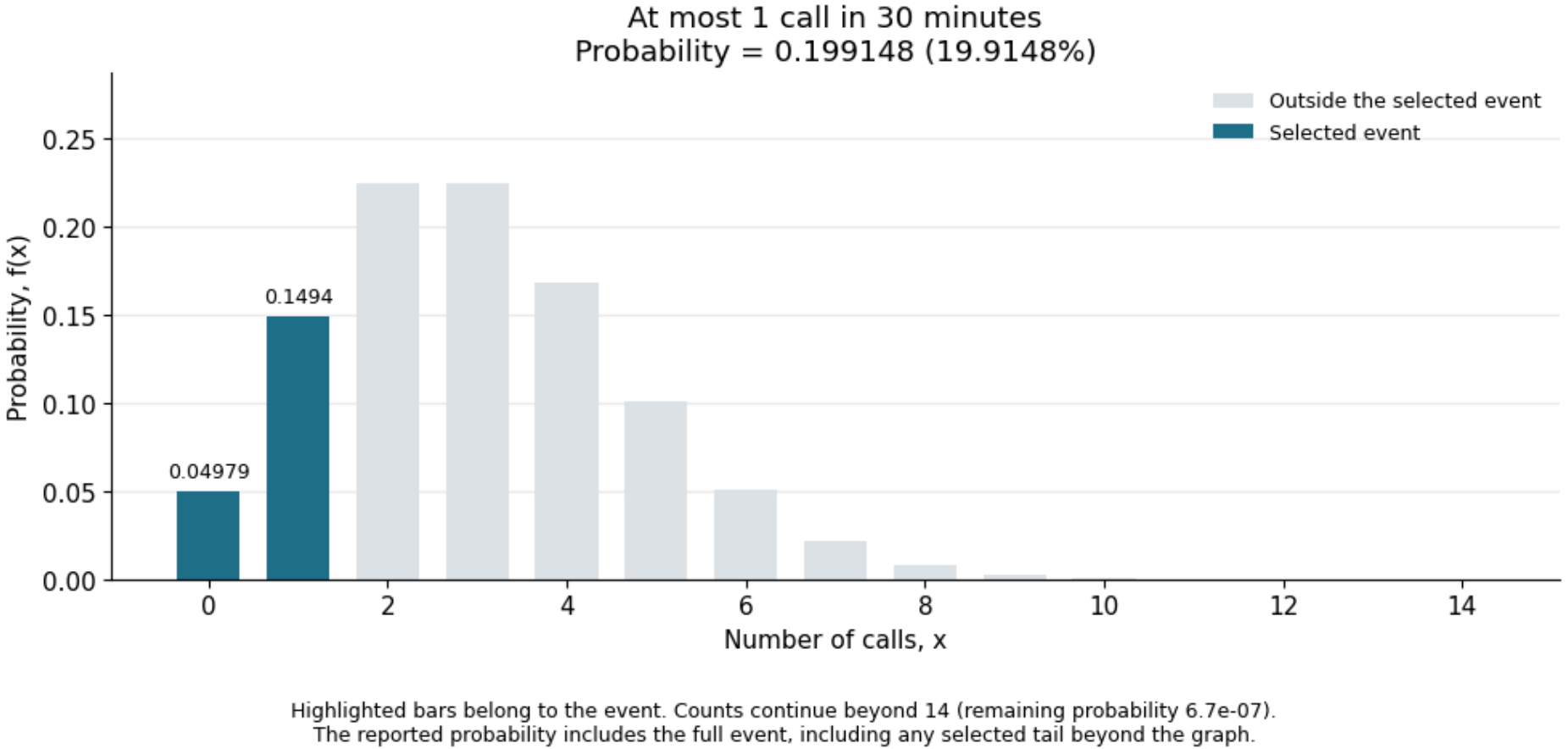
$$P(X \le 1) = \sum_{x=0}^{1} f(x).$$

Calls, x	Calculation of $f(x)$	Probability
0	$\frac{3^0 e^{-3}}{0!}$	0.0497871
1	$\frac{3^1 e^{-3}}{1!}$	0.149361

$$P(X \le 1) \approx 0.0497871 + 0.149361 \approx 0.199148 = 19.9148\%.$$

Results use unrounded probabilities.

Poisson Probabilities: Length



Highlighted bars belong to the event. Counts continue beyond 14 (remaining probability 6.7e-07).
The reported probability includes the full event, including any selected tail beyond the graph.

Poisson Mean and Variance

For $X \sim \text{Poisson}(\mu)$,

$$E(X) = \mu, \qquad \text{Var}(X) = \mu, \qquad \sigma = \sqrt{\mu}.$$

- Mean and variance: same numerical value, different units

For Beta Support:

Interval	Mean (calls)	Variance (calls squared)	SD (calls)
10 minutes	1	1	1
30 minutes	3	3	$\sqrt{3} \approx 1.732$
60 minutes	6	6	$\sqrt{6} \approx 2.449$

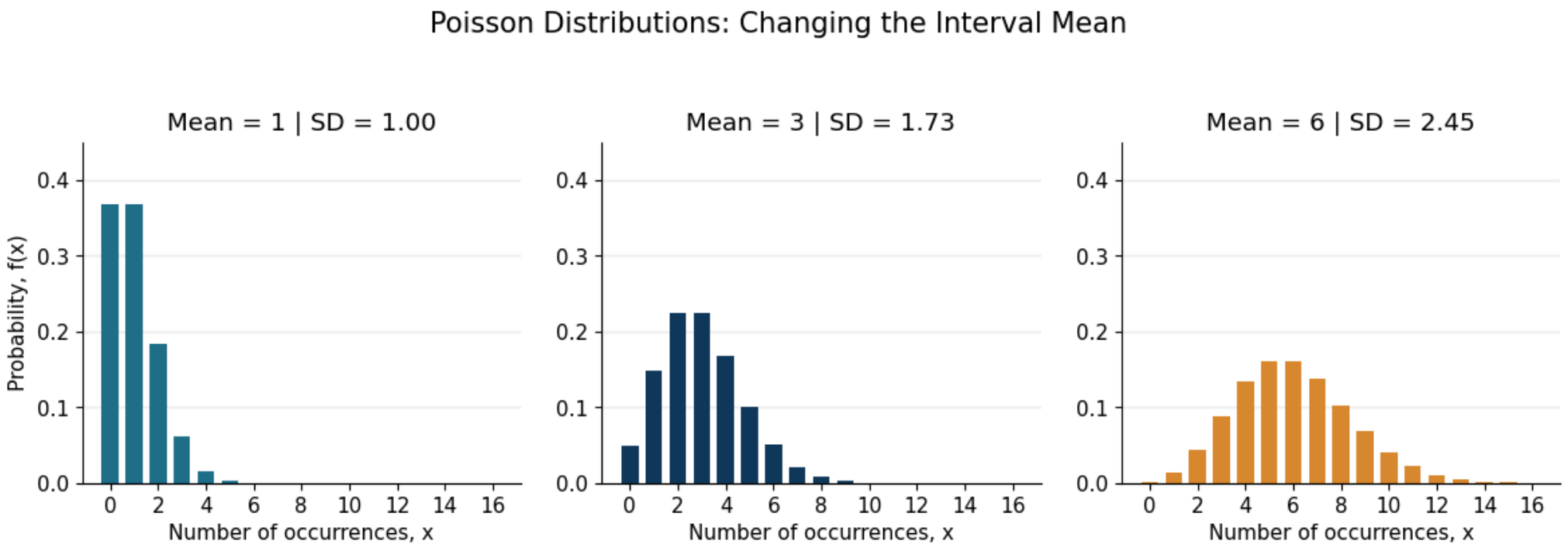
From 30 to 60 minutes:

- Mean and variance: double
- Standard deviation: multiplied by $\sqrt{2}$

$$\frac{\sqrt{6}}{\sqrt{3}} = \sqrt{2}.$$

Equality of the theoretical mean and variance is a property of the Poisson model, not a rule that every observed set of counts must satisfy exactly.

Poisson Mean and Variance



Exercise 4

Calls reach Delta Reservations according to a Poisson process with a constant rate.

	Average arrival rate
	12 calls per hour

a. Find the probability that at most one call arrives during a ten-minute interval.

b. Find the probability that each of two consecutive five-minute intervals contains at least one call.

c. Find the probability that at least one call arrives during the full ten minutes. Explain why this differs from part (b).

Exercise 4: Solution

For ten minutes,

$$\mu_{10} = 12\left(\frac{10}{60}\right) = 2.$$

Thus,

$$P(X_{10} \leq 1) = e^{-2} + 2e^{-2} = 3e^{-2} \approx 0.406006.$$

For each five-minute interval,

$$\mu_5 = 12\left(\frac{5}{60}\right) = 1, \qquad P(X_5 \geq 1) = 1 - e^{-1} \approx 0.632121.$$

Let A and B be the call counts in the first and second five-minute intervals. These intervals do not overlap, so their counts are independent:

$$P(A \geq 1,\ B \geq 1) = (1 - e^{-1})^2 \approx 0.399576.$$

For part (c), the total number of calls is $A + B$. Its complement is no calls during the full ten minutes:

$$P(A + B \geq 1) = 1 - P(A + B = 0).$$

Since call counts cannot be negative, $A + B = 0$ requires $A = 0$ and $B = 0$. For each five-minute interval,

$$P(A = 0) = P(B = 0) = \frac{1^0 e^{-1}}{0!} = e^{-1}.$$

Exercise 4: Solution

Using independence,

$$P(A+B=0) = P(A=0,\ B=0) = P(A=0)P(B=0) = e^{-1}e^{-1} = e^{-2}.$$

Therefore,

$$P(A+B \geq 1) = 1 - e^{-2} = 1 - 0.135335\ldots \approx 0.864665.$$

- Part (b): at least one call in both intervals
- Part (c): at least one call in either interval or both; also includes $A \geq 1,\ B = 0$ and $A = 0,\ B \geq 1$

Sampling Without Replacement

Beta Services:

- Team of ten: four certified employees and six non-certified employees
- Random selection of three different employees
- Every group of three equally likely

Let X count certified employees in the selected group.

Selection	Probability of a certified employee
First employee	$4/10$
Second, if the first is certified	$3/9$
Second, if the first is not certified	$4/9$

- Conditional probability changes with the previous selection.
- Selections are not independent.

For example, the probability that the first two are certified is

$$\frac{4}{10}\frac{3}{9} = \frac{2}{15} \approx 0.133333,$$

not $(4/10)^2 = 0.16$. A binomial model is not exact for this sampling procedure.

Hypergeometric Probability Distribution

A hypergeometric random variable counts successes in a simple random sample drawn without replacement from a finite population.

Symbol	Meaning	Beta Services
N	Population size	10 employees
r	Successes in the population	4 certified employees
n	Sample size	3 employees
x	Successes in the sample	$0, 1, 2, 3$

A sample cannot contain more successes or failures than the population contains:

$$x \le r, \qquad n - x \le N - r.$$

Together with $0 \le x \le n$, this gives the possible integer values

$$\max(0, n - (N - r)) \le x \le \min(n, r).$$

Now suppose the team has seven certified employees instead of four, and five employees are selected:

Certified employees, r	Non-certified employees, $N - r$	Employees selected, n
7	$10 - 7 = 3$	5

- Fewest certified: select all three non-certified employees. The remaining $5 - 3 = 2$ must be certified.
- Most certified: all five selected employees can be certified because seven are available.

Hypergeometric Probability Distribution

$$x_{\min} = \max(0, 5 - 3) = 2, \qquad x_{\max} = \min(5, 7) = 5.$$

Therefore, the possible counts are $x = 2, 3, 4, 5$. Selecting zero or one certified employee would require more than the three non-certified employees available.

Hypergeometric Probability Function

To select exactly x successes in a sample of n:

Selection	Number of ways
Choose x successes from the r available	$\binom{r}{x}$
Choose the remaining $n-x$ elements from the $N-r$ failures	$\binom{N-r}{n-x}$
Choose any sample of n from all N elements	$\binom{N}{n}$

- Pair each choice of successes with each choice of failures.
- All samples of size n are equally likely.

$$f(x) = P(X = x) = \frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}.$$

For exactly one certified employee in Beta's group:

$$P(X = 1) = \frac{\binom{4}{1}\binom{6}{2}}{\binom{10}{3}} = \frac{4(15)}{120} = 0.50.$$

Order does not matter: the sample is a group, not a sequence of appointments.

Hypergeometric Probabilities: Employees

For Beta Services, $N = 10$, $r = 4$, and $n = 3$.

$$\binom{10}{3} = \frac{10!}{3!7!} = \frac{10(9)(8)}{3(2)(1)} = 120.$$

Certified employees, x	Ways to select certified employees	Ways to select others	$f(x)$
0	$\binom{4}{0} = 1$	$\binom{6}{3} = 20$	$20/120 = 1/6$
1	$\binom{4}{1} = 4$	$\binom{6}{2} = 15$	$60/120 = 1/2$
2	$\binom{4}{2} = 6$	$\binom{6}{1} = 6$	$36/120 = 3/10$
3	$\binom{4}{3} = 4$	$\binom{6}{0} = 1$	$4/120 = 1/30$

$$\sum_{x=0}^{3} f(x) = \frac{20 + 60 + 36 + 4}{120} = 1.$$

- 120 possible groups: equally likely
- Four possible counts: not equally likely

Hypergeometric Probabilities: Events

Continue with the three employees selected from Beta's team.

- At least one certified employee:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{20}{120} = \frac{5}{6} \approx 0.833333.$$

- More than one certified employee:

$$P(X > 1) = P(X = 2) + P(X = 3) = \frac{36 + 4}{120} = \frac{1}{3} \approx 0.333333.$$

- Between one and two certified employees, inclusive:

$$P(1 \leq X \leq 2) = \frac{60 + 36}{120} = 0.80.$$

If the group must include both certified and non-certified employees, its probability is also 0.80: exclude both $X = 0$ and $X = 3$.

Hypergeometric Probabilities: Events

VBox(children=(Dropdown(description='Event:', index=3, layout=Layout(max_width='100%', width='650px'), options…

Hypergeometric Probabilities: Employees

Beta Services: select 3 different employees at random from 10, of whom 4 are certified. No replacement.

$$N = 10, \quad r = 4, \quad n = 3, \qquad f(x) = \frac{\binom{4}{x}\binom{6}{3-x}}{\binom{10}{3}}.$$

At least 1 certified employee.

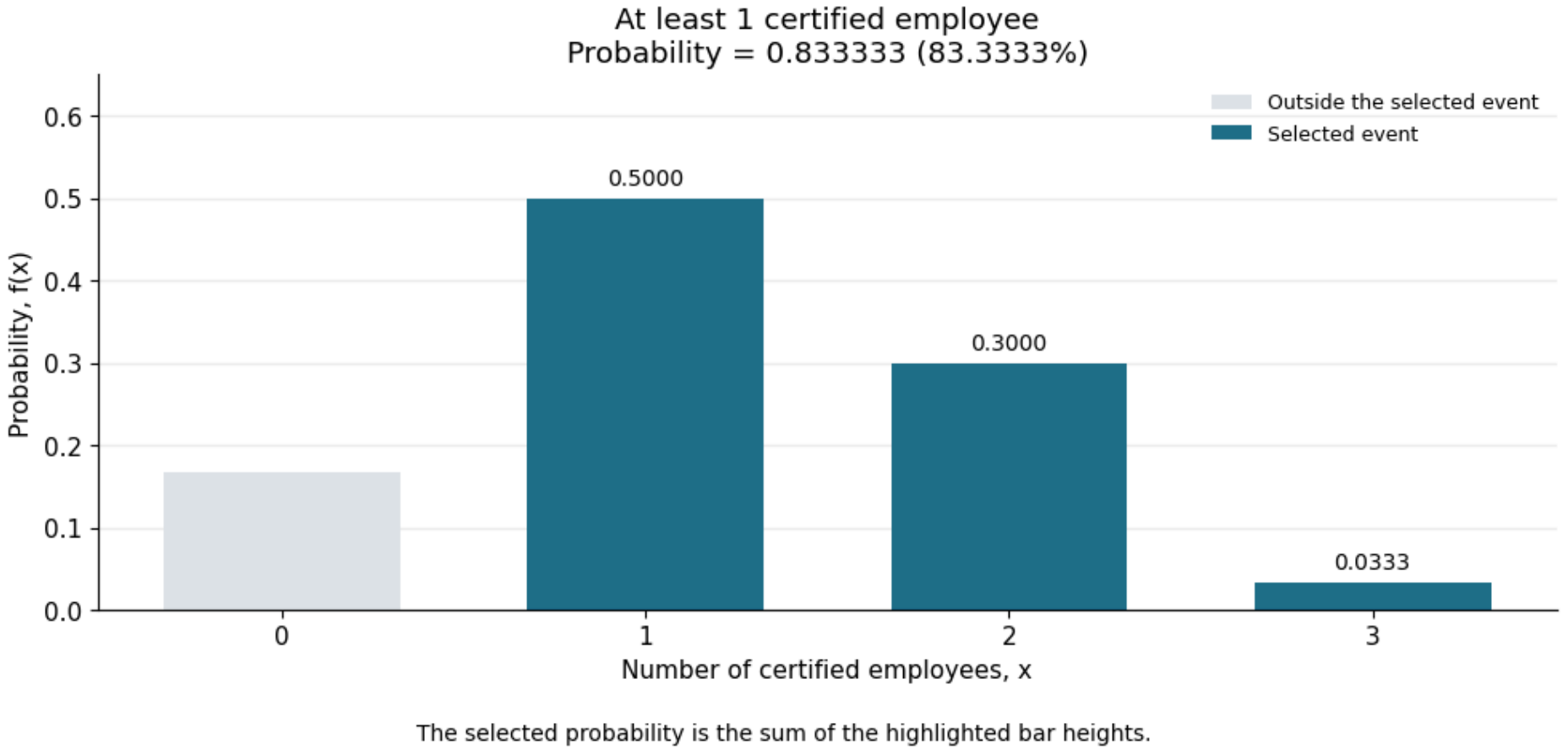
$$P(X \geq 1) = 1 - [f(0)]\,.$$

Subtract the probabilities of the excluded counts from one.

Certified employees, x	Calculation of $f(x)$	Probability
0	$\frac{\binom{4}{0}\binom{6}{3}}{\binom{10}{3}} = \frac{1(20)}{120} = \frac{20}{120}$	0.166667

$$P(X \geq 1) = 1 - \left[\frac{20}{120}\right] = \frac{5}{6} \approx 0.833333 \quad (83.3333\%).$$

Hypergeometric Probabilities: Events



The selected probability is the sum of the highlighted bar heights.

Hypergeometric Mean and Variance

For a sample of n from a population of $N > 1$ containing r successes, let $p = r/N$.

$$E(X) = np = n\frac{r}{N}.$$

$$\mathrm{Var}(X) = np(1-p)\frac{N-n}{N-1}, \qquad \sigma = \sqrt{np(1-p)\frac{N-n}{N-1}}.$$

- $(N-n)/(N-1)$: correction for sampling without replacement
- For $1 < n \leq N$ and $0 < r < N$: variance below the corresponding binomial variance

For Beta Services:

$$\mu = 3\left(\frac{4}{10}\right) = 1.20 \text{ certified employees},$$

$$\sigma^2 = 3\left(\frac{4}{10}\right)\left(\frac{6}{10}\right)\left(\frac{10-3}{10-1}\right) = 0.56 \text{ employees}^2,$$

$$\sigma = \sqrt{0.56} \approx 0.7483 \text{ employees}.$$

If the whole population is selected, $n = N$ and $X = r$ with certainty; the variance is zero.

Hypergeometric Mean and Variance

Beta Services: direct calculation from the complete probability distribution.

x	$f(x)$	$xf(x)$	$x^2 f(x)$
0	1/6	0	0
1	1/2	1/2	1/2
2	3/10	6/10	12/10
3	1/30	3/30	9/30
Total	1	1.20	2.00

$$E(X) = 0 + \frac{1}{2} + \frac{6}{10} + \frac{3}{30} = 1.20.$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 2.00 - (1.20)^2 = 0.56.$$

- Mean of 1.20: long-run average across repeated random groups
- Not a possible count in a single group

Sampling With and Without Replacement

Consider three random draws from ten items, four of which are labelled success.

Feature	Replace and mix after every draw	Do not replace
Model	Binomial, $n = 3,\ p = 0.40$	Hypergeometric, $N = 10,\ r = 4,\ n = 3$
Success probability after a success	$4/10$	$3/9$
Mean	$3(0.40) = 1.20$	$3(4/10) = 1.20$
Variance	$3(0.40)(0.60) = 0.72$	$0.72(7/9) = 0.56$

- With replacement: independent draws; the same item can appear again.
- Without replacement: three different items selected.

$$P(X = 0) \text{ with replacement} = (0.60)^3 = 0.216.$$

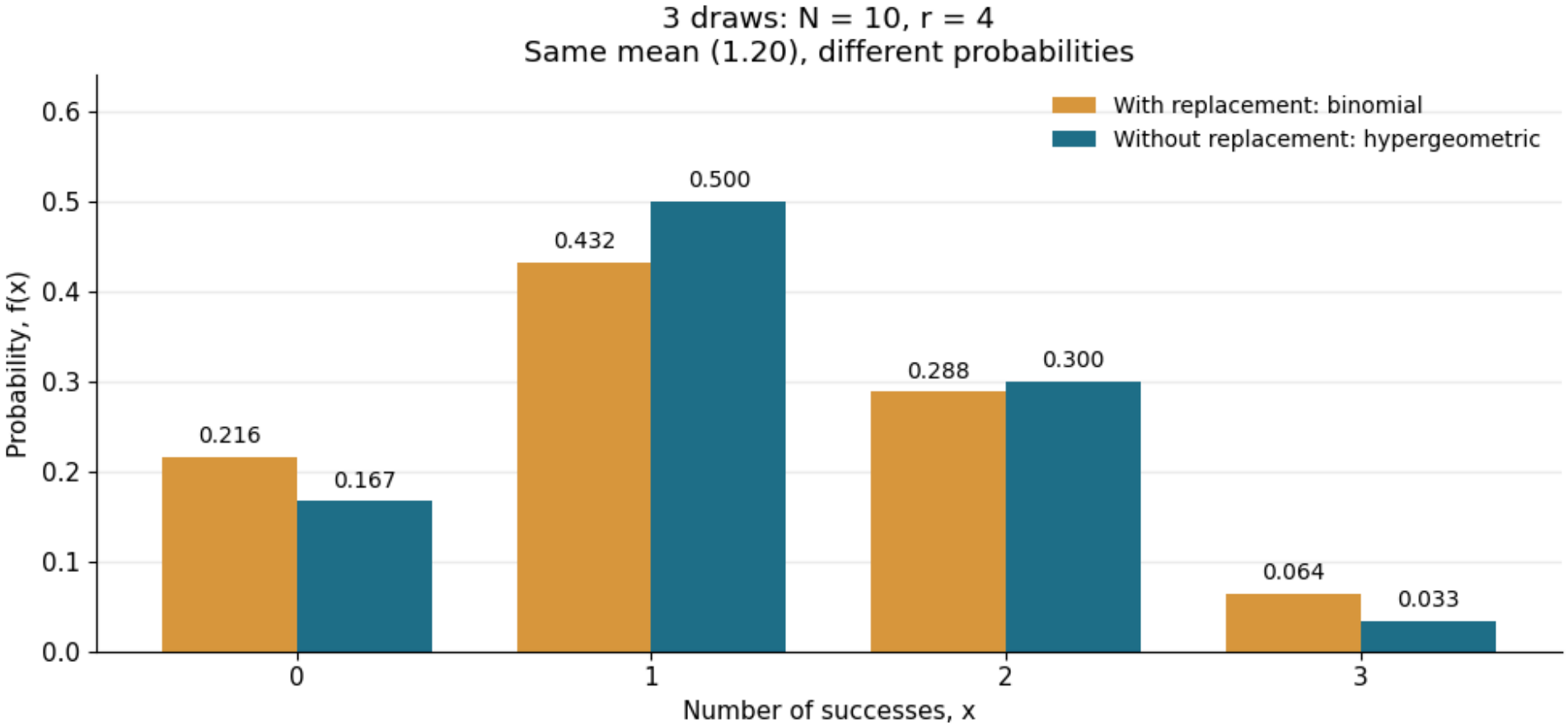
$$P(X = 0) \text{ without replacement} = \frac{6}{10}\frac{5}{9}\frac{4}{8} = \frac{\binom{6}{3}}{\binom{10}{3}} = \frac{1}{6} \approx 0.166667.$$

Equal means do not make the distributions interchangeable.

Sampling With and Without Replacement

VBox(children=(BoundedIntText(value=3, description='Draws (n):', max=10, min=1, style=DescriptionStyle(descrip…

The mean is $\mu = nr/N$. Selecting a mean changes r; the menu lists only means possible for the current n and N.



Binomial Approximation to the Hypergeometric Distribution

- Small sample relative to the population: little change in success probability as items are removed.
- Binomial model with $p = r/N$: an approximation to the hypergeometric model under this condition.

Keep $n = 3$ and $r/N = 0.40$:

Population, N	Successes, r	Sample fraction, n/N	Exact $P(X = 0)$	Binomial $P(X = 0)$
10	4	30%	0.166667	0.216000
100	40	3%	0.211626	0.216000
1,000	400	0.3%	0.215567	0.216000

The hypergeometric probability formula is:

$$P_{\text{Hyp}}(X = x) = \frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}.$$

- Here $n = 3$ and $x = 0$: select no successes from the r successes, and all three items from the $N - r$ items that are not successes.

$$P_{\text{Hyp}}(X = 0) = \frac{\binom{r}{0}\binom{N-r}{3-0}}{\binom{N}{3}} = \frac{1 \times \binom{N-r}{3}}{\binom{N}{3}}.$$

Binomial Approximation to the Hypergeometric Distribution

- $N = 10$, $r = 4$:

$$P_{\text{Hyp}}(X = 0) = \frac{\binom{4}{0}\binom{10-4}{3-0}}{\binom{10}{3}} = \frac{1 \times 20}{120} \approx 0.166667.$$

- $N = 100$, $r = 40$:

$$P_{\text{Hyp}}(X = 0) = \frac{\binom{40}{0}\binom{100-40}{3-0}}{\binom{100}{3}} = \frac{1 \times 34{,}220}{161{,}700} \approx 0.211626.$$

- $N = 1000$, $r = 400$:

$$P_{\text{Hyp}}(X = 0) = \frac{\binom{400}{0}\binom{1000-400}{3-0}}{\binom{1000}{3}} = \frac{1 \times 35{,}820{,}200}{166{,}167{,}000} \approx 0.215567.$$

- Binomial approximation: use the same probability of no success, $1 - p = 0.60$, on each draw.

$$P_{\text{Bin}}(X = 0) = (1 - p)^3 = (0.60)^3 = 0.216.$$

- The binomial probability stays at 0.216 because n and p are unchanged. As n/N decreases, the exact probability moves closer to it.

Compare the variances:

$$\sigma^2_{\text{Bin}} = np(1 - p) = 3(0.40)(0.60) = 0.72,$$

Binomial Approximation to the Hypergeometric Distribution

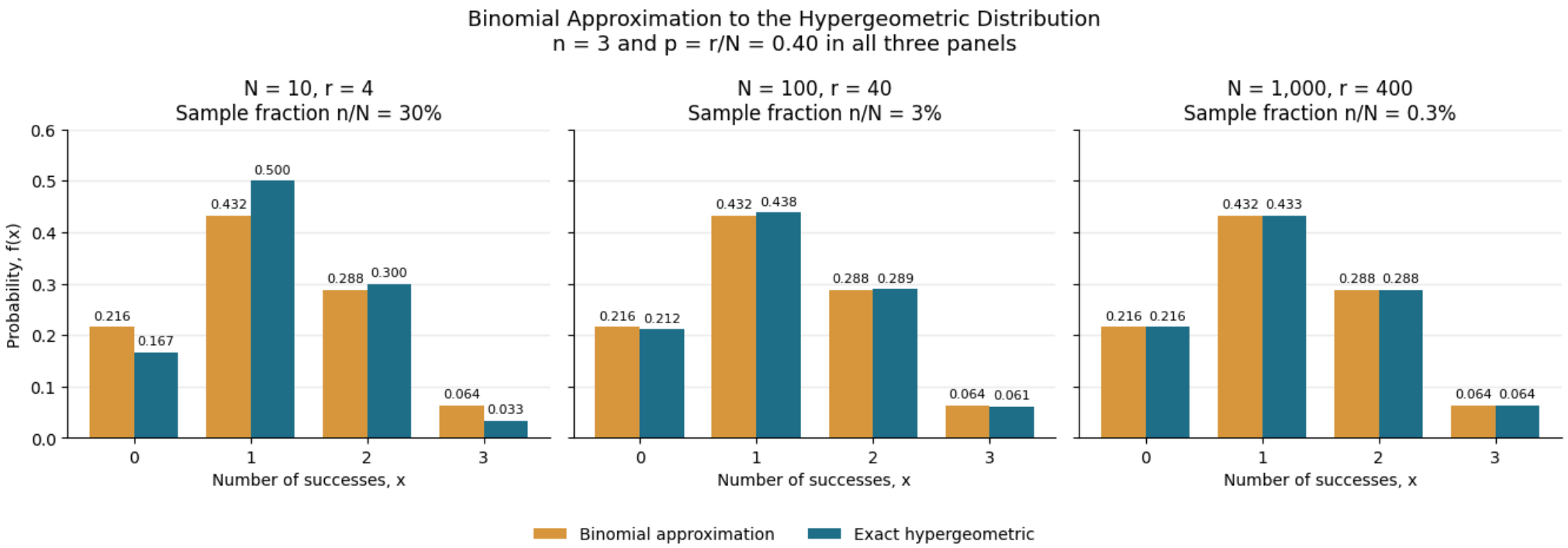
$$\sigma^2_{\text{Hyp}} = \sigma^2_{\text{Bin}} \frac{N-n}{N-1} = 0.72 \frac{N-3}{N-1}.$$

The factor multiplying the binomial variance approaches 1:

$$\frac{10-3}{10-1} \approx 0.7778, \qquad \frac{100-3}{100-1} \approx 0.9798, \qquad \frac{1000-3}{1000-1} \approx 0.9980.$$

- For $N = 10$, the exact variance is $0.72(7/9) = 0.56$. For $N = 1000$, it is approximately 0.7186, almost the binomial variance of 0.72.
- A large population is not enough: the sample must be small relative to it.
- Use the exact model when population composition is given and the calculation is manageable.

Binomial Approximation to the Hypergeometric Distribution



Exercise 5

Gamma Electronics evaluates an inspection plan for a lot containing exactly three defective items.

- Random sampling without replacement
- Every sample of the chosen size equally likely

Lot size	Defective items	Candidate sample sizes
12	3	2, 3, or 4 items

The lot is rejected if the sample contains at least one defective item.

a. Choose the smallest candidate sample size that gives at least a 70% probability of rejecting this lot. Show the probabilities for all three candidates.

b. For the chosen sample size, calculate the mean and standard deviation of the number of defective items found.

c. What does an accepted sample establish about the uninspected items in this particular lot?

Exercise 5: Solution

Let X be the number of defective items in the sample. Here $N = 12$ and $r = 3$.

$$P(X = x) = \frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}.$$

For sample size n, the lot is accepted only when all sampled items are among its nine non-defective items.

$$P(\text{reject}) = 1 - P(X = 0) = 1 - \frac{\binom{3}{0}\binom{12-3}{n-0}}{\binom{12}{n}} = 1 - \frac{\binom{3}{0}\binom{9}{n}}{\binom{12}{n}}.$$

n	$P(X = 0)$	$P(\text{reject})$
2	$\frac{\binom{3}{0}\binom{9}{2}}{\binom{12}{2}} = \frac{1 \times 36}{66} \approx 0.545455$	$1 - \frac{36}{66} \approx 0.454545$
3	$\frac{\binom{3}{0}\binom{9}{3}}{\binom{12}{3}} = \frac{1 \times 84}{220} \approx 0.381818$	$1 - \frac{84}{220} \approx 0.618182$
4	$\frac{\binom{3}{0}\binom{9}{4}}{\binom{12}{4}} = \frac{1 \times 126}{495} \approx 0.254545$	$1 - \frac{126}{495} \approx 0.745455$

Only $n = 4$ meets the 70% requirement.

$$E(X) = n\frac{r}{N} = 4\left(\frac{3}{12}\right) = 1 \text{ defective item.}$$

Exercise 5: Solution

$$\begin{aligned}\mathrm{Var}(X) &= n\frac{r}{N}\left(1-\frac{r}{N}\right)\frac{N-n}{N-1}\\ &= 4\left(\frac{3}{12}\right)\left(1-\frac{3}{12}\right)\left(\frac{12-4}{12-1}\right)\\ &= 4\left(\frac{3}{12}\right)\left(\frac{9}{12}\right)\left(\frac{8}{11}\right)=\frac{6}{11}\approx 0.545455.\end{aligned}$$

$$\sigma=\sqrt{\mathrm{Var}(X)}=\sqrt{4\left(\frac{3}{12}\right)\left(\frac{9}{12}\right)\left(\frac{8}{11}\right)}=\sqrt{\frac{6}{11}}\approx 0.7385 \text{ items.}$$

If the sample is accepted:

- No defective item selected
- All three defects remain among the eight uninspected items.
- Acceptance does not establish a defect-free lot.

Probability Distributions: Comparison

Feature	Binomial	Poisson	Hypergeometric
What is counted?	Successes in n trials	Occurrences in an interval	Successes in a sample of n
Parameters	n, p	Expected interval count, μ	N, r, n
Key conditions	Independent trials, common p	Constant rate, independent non-overlapping counts, individual occurrences	Random sampling without replacement from a fixed population
Possible values	$0, \dots, n$	$0, 1, 2, \dots$	$\max(0, n - N + r), \dots, \min(n, r)$
Mean	np	μ	nr/N
Variance	$np(1-p)$	μ	$n(r/N)(1 - r/N)(N - n)/(N - 1)$

- Model choice: the random variable and the sampling or arrival process
- Different counting problems:
  - Defects among a fixed number of independently produced items
  - Flaws along a length of cable

Probabilities in Excel

Last argument:

- FALSE: exact probability
- TRUE: cumulative probability $P(X \le k)$

Model and parameters	Exact probability $P(X = 1)$	At most one, $P(X \le 1)$
Binomial: $n = 5,\ p = 0.40$	`=BINOM.DIST(1,5,0.4,FALSE)`	`=BINOM.DIST(1,5,0.4,TRUE)`
Poisson: $\mu = 3$	`=POISSON.DIST(1,3,FALSE)`	`=POISSON.DIST(1,3,TRUE)`
Hypergeometric: $N = 10,\ r = 4,\ n = 3$	`=HYPGEOM.DIST(1,3,4,10,FALSE)`	`=HYPGEOM.DIST(1,3,4,10,TRUE)`

Results:

- Binomial: $(0.259200,\ 0.336960)$
- Poisson: $(0.149361,\ 0.199148)$
- Hypergeometric: $(0.500000,\ 0.666667)$

For at least k, subtract the cumulative probability through $k - 1$ from one. For example,

```
=1-HYPGEOM.DIST(1,3,4,10,TRUE)
```

gives $P(X \ge 2) = 1 - 0.666667 = 0.333333$.

- Argument separator: commas or semicolons, depending on the Excel installation
- POISSON.DIST: interval mean, not the hourly rate

Exercise 6

Consider three counts recorded by Delta Services.

Record	Process	Count of interest
A	Four customers decide independently; each buys with probability 0.20.	Customers who buy
B	Four different orders are randomly selected from 20 orders, exactly four of which require correction.	Selected orders requiring correction
C	Calls arrive at a constant rate of four per hour according to a Poisson process.	Calls in 30 minutes

For each record, identify the distribution and its parameters, calculate the expected count, and find the probability of a count of at least two.

Could Records A and B use the same exact probability calculation because both involve four observations and a population proportion or success probability of 0.20? Explain using their selection processes.

Exercise 6: Solution

Record	Model	Expected count
A	Binomial, $n = 4,\ p = 0.20$	$4(0.20) = 0.80$ buyers
B	Hypergeometric, $N = 20,\ r = 4,\ n = 4$	$4(4/20) = 0.80$ orders
C	Poisson, $\mu = 4(30/60) = 2$	2 calls

For A:

$$P(X_A \geq 2) = 1 - (0.80)^4 - 4(0.20)(0.80)^3 = 1 - 0.4096 - 0.4096 = 0.1808.$$

For B:

$$P(X_B \geq 2) = 1 - \frac{\binom{4}{0}\binom{16}{4} + \binom{4}{1}\binom{16}{3}}{\binom{20}{4}}$$

$$= 1 - \frac{1820 + 4(560)}{4845} = \frac{785}{4845} \approx 0.162023.$$

For C:

$$P(X_C \geq 2) = 1 - e^{-2} - 2e^{-2} \approx 0.593994.$$

Exercise 6: Solution

- A: independent decisions with constant p
- B: sampling without replacement
  - After selecting an order requiring correction: next probability $3/19$, not 0.20
- Equal means do not imply equal probabilities.

What is next?

- Week 7: Continuous Probability Distributions
  - Continuous Random Variables and Probability Density
  - Uniform Probability Distribution
  - Normal Probability Distribution
  - Student's t-Distribution: Introduction
  - Normal Approximation of Binomial Probabilities
  - Exponential Probability Distribution

Exercise 7

Epsilon Couriers receives collection requests according to a Poisson process.

- Requests in each booking window: collected together after the window closes
- No requests carried over from earlier windows

Arrival rate	Booking window
8 requests per hour	15 minutes

Before a window begins, the firm chooses one vehicle:

Vehicle capacity (requests)	Vehicle fee per window (MAD)
3	20
4	27
5	36

- Requests above capacity: all excess requests collected by an external provider
- One fixed additional fee of 50 MAD, regardless of the number of excess requests
- No other costs

Choose the vehicle with the lowest expected total cost among those requiring the external provider in no more than 10% of windows. Support the choice numerically. Would the choice change if the limit were 15%?

Exercise 7: Solution

Let X count requests in one 15-minute booking window.

$$\mu = 8\left(\frac{15}{60}\right) = 2, \qquad X \sim \text{Poisson}(2), \qquad f(x) = \frac{2^x e^{-2}}{x!}.$$

Requests, x	0	1	2	3	4	5
$f(x)$	0.135335	0.270671	0.270671	0.180447	0.090224	0.036089

For example, $f(3) = 2^3 e^{-2}/3! \approx 0.180447$.

A vehicle of capacity c handles all requests when $X \le c$. The external provider is needed when $X > c$, meaning $X \ge c + 1$.

Add the probabilities of the counts that the vehicle can handle:

$$P(X \le 3) = f(0) + f(1) + f(2) + f(3) \approx 0.857123.$$

$$P(X \le 4) = P(X \le 3) + f(4) \approx 0.947347.$$

$$P(X \le 5) = P(X \le 4) + f(5) \approx 0.983436.$$

Exercise 7: Solution

Subtract each cumulative probability from one:

Capacity	External provider needed when	Probability
3	$X \geq 4$	$1 - P(X \leq 3) \approx 1 - 0.857123 = 0.142877$
4	$X \geq 5$	$1 - P(X \leq 4) \approx 1 - 0.947347 = 0.052653$
5	$X \geq 6$	$1 - P(X \leq 5) \approx 1 - 0.983436 = 0.016564$

Cumulative probabilities use unrounded $f(x)$ values.

Exercise 7: Solution

The vehicle fee is paid in every window. The additional 50 MAD is paid once if requests exceed capacity, not once per excess request.

Capacity	Cost when $X \le c$	Cost when $X > c$
3	20 MAD	$20 + 50 = 70$ MAD
4	27 MAD	$27 + 50 = 77$ MAD
5	36 MAD	$36 + 50 = 86$ MAD

Weight each possible cost by its probability:

$$E(C_3) \approx 20(0.857123) + 70(0.142877) \approx 27.14 \text{ MAD}.$$

$$E(C_4) \approx 27(0.947347) + 77(0.052653) \approx 29.63 \text{ MAD}.$$

$$E(C_5) \approx 36(0.983436) + 86(0.016564) \approx 36.83 \text{ MAD}.$$

In general, let F_c be the vehicle fee and C_c the total cost:

$$E(C_c) = F_c P(X \le c) + (F_c + 50)P(X > c).$$

$$E(C_c) = F_c[P(X \le c) + P(X > c)] + 50P(X > c) = F_c + 50P(X > c).$$

First apply the probability limit, then compare expected costs among the qualifying vehicles.

Exercise 7: Solution

- 10% limit:
  - Capacity 3 is excluded: $14.2877\% > 10\%$, despite its lower expected cost.
  - Capacities 4 and 5 qualify: 5.2653% and 1.6564%.
  - Choose capacity 4: $29.63 < 36.83$ MAD.
- 15% limit:
  - All three qualify, including capacity 3 with $14.2877\% < 15\%$.
  - Choose capacity 3: $27.14 < 29.63 < 36.83$ MAD.

Exercise 8

Epsilon sells five one-year service contracts.

- At most one claim per contract
- Independent claims across contracts, with the same probability for each

Contracts sold	Claim probability per contract	Payment per claim
5	0.10	200 MAD

- Same premium per contract; all premiums received at the start
- No other revenues or costs
- Profit: total premiums minus total claim payments
- Breaking even is not a loss.

a. Find the smallest whole-MAD premium per contract that makes the probability of a loss no greater than 1%.

b. At that premium, calculate expected total profit and its standard deviation.

Exercise 8: Solution

Let X count claims and a be the premium per contract.

$$X \sim \text{Bin}(5, 0.10), \qquad Y = 5a - 200X.$$

A loss occurs when

$$5a - 200X < 0 \quad \Longleftrightarrow \quad X > \frac{a}{40}.$$

Calculate the probabilities needed to locate the smallest acceptable premium:

Claims, x	Calculation	$f(x)$
0	$(0.90)^5$	0.590490
1	$5(0.10)(0.90)^4$	0.328050
2	$10(0.10)^2(0.90)^3$	0.072900

$$P(X \geq 2) = 1 - 0.590490 - 0.328050 = 0.081460 > 0.01.$$

$$P(X \geq 3) = 1 - 0.590490 - 0.328050 - 0.072900 = 0.008560 < 0.01.$$

The premiums must cover two claims without a loss:

$$5a \geq 2(200) \quad \Longrightarrow \quad a \geq 80 \text{ MAD}.$$

Exercise 8: Solution

- At $a = 80$: two claims give zero profit; a loss requires at least three claims.
- Below 80 MAD: two or more claims cause a loss, so the loss probability is at least 8.146%. Still lower premiums can also produce a loss with fewer claims.
- Smallest acceptable premium: 80 MAD

Exercise 8: Solution

At 80 MAD per contract,

$$Y = 5(80) - 200X = 400 - 200X.$$

For the number of claims:

$$E(X) = 5(0.10) = 0.50,$$

$$\mathrm{Var}(X) = 5(0.10)(0.90) = 0.45.$$

Therefore,

$$E(Y) = 400 - 200(0.50) = 300 \text{ MAD},$$

$$\mathrm{Var}(Y) = (-200)^2(0.45) = 18{,}000 \text{ MAD}^2,$$

$$\sigma_Y = \sqrt{18{,}000} \approx 134.16 \text{ MAD}.$$

- Loss probability: 0.856%, not zero
- Expected profit does not limit the loss in a particular year.

Exercise 9

Zeta Support: a randomly selected 20-minute interval.

- One condition throughout the interval, initially unobserved
- Conditional on that condition: a Poisson process at the stated rate

Condition	Probability	Call rate
Regular	0.60	3 per hour
Busy	0.40	9 per hour

a. Find the probability of at least three calls in the interval.

b. The interval contains at least three calls, but the exact count is not reported. Find the probability that the condition was busy.

c. An analyst averages the two rates using the condition probabilities and uses that rate in a single Poisson model. Calculate the resulting probability of the event in part (a). Is it the same model?

Exercise 9: Solution

Let X count calls and $A = \{X \geq 3\}$. Let R and B denote the regular and busy conditions. The 20-minute means are

$$\mu_R = 3\left(\frac{20}{60}\right) = 1, \qquad \mu_B = 9\left(\frac{20}{60}\right) = 3.$$

Under the regular condition:

$$P(A \mid R) = 1 - [P(X = 0 \mid R) + P(X = 1 \mid R) + P(X = 2 \mid R)].$$

$$P(A \mid R) = 1 - \left(e^{-1} + e^{-1} + \frac{e^{-1}}{2}\right) = 1 - 2.5e^{-1} \approx 0.080301.$$

Under the busy condition:

$$P(A \mid B) = 1 - [P(X = 0 \mid B) + P(X = 1 \mid B) + P(X = 2 \mid B)].$$

$$P(A \mid B) = 1 - \left(e^{-3} + 3e^{-3} + \frac{9e^{-3}}{2}\right) = 1 - 8.5e^{-3} \approx 0.576810.$$

The unconditional probability is

$$P(A) = 0.60P(A \mid R) + 0.40P(A \mid B)$$

$$\approx 0.60(0.080301) + 0.40(0.576810) = 0.278905.$$

The busy condition and at least three calls occur together with probability

Exercise 9: Solution

$$P(B \cap A) = 0.40 P(A \mid B) \approx 0.230724.$$

Thus,

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} \approx \frac{0.230724}{0.278905} = 0.827250.$$

- Busy-condition probability: 40% before the evidence; about 82.72% afterward

Results use unrounded probabilities.

Exercise 9: Solution

The analyst's weighted average rate is

$$\bar{\lambda} = 0.60(3) + 0.40(9) = 5.4 \text{ calls per hour.}$$

The corresponding 20-minute mean is

$$\bar{\mu} = 5.4\left(\frac{20}{60}\right) = 1.8.$$

A single Poisson model with this mean gives

$$P(X \geq 3) = 1 - e^{-1.8}\left(1 + 1.8 + \frac{1.8^2}{2}\right) \approx 0.269379,$$

rather than the correct mixture probability, 0.278905.

Both calculations have the same expected count:

$$E(X) = 0.60(1) + 0.40(3) = 1.8.$$

- Actual interval rate: either three or nine, not 5.4
- Averaging parameters does not average probabilities.
- Same mean does not establish the same distribution.

Exercise 10

Eta Imports receives lots of ten items. Supplier records establish the following lot composition before inspection:

Lot type	Proportion of incoming lots	Defective items in the lot
A	0.70	1
B	0.30	4

- Individual lot type: unknown
- Choice: randomly inspect three or four different items without replacement.
- Every sample of the chosen size equally likely
- Tests: perfect defect identification; no damage to items

The lot is accepted only if no defect is found.

Cost	Amount
Inspection	6 MAD per sampled item
Additional charge if a Type B lot is accepted	120 MAD per lot
Other costs for this comparison	0 MAD

Recommend one inspection plan. Support your recommendation with the acceptance probability, the probability that an accepted lot is Type B, and expected total cost per incoming lot. Explain the trade-off behind your choice.

Exercise 10: Solution

Let G denote acceptance. All n sampled items must be nondefective.

- Type A: $10 - 1 = 9$ nondefective items; $\binom{9}{n}$ samples contain no defects.
- Type B: $10 - 4 = 6$ nondefective items; $\binom{6}{n}$ samples contain no defects.
- Each lot has $\binom{10}{n}$ equally likely samples of size n.

$$P(G \mid A) = \frac{\binom{9}{n}}{\binom{10}{n}}, \qquad P(G \mid B) = \frac{\binom{6}{n}}{\binom{10}{n}}.$$

Sample size	$P(G \mid A)$	$P(G \mid B)$
3	$84/120 = 0.700000$	$20/120 = 0.166667$
4	$126/210 = 0.600000$	$15/210 = 0.071429$

For three inspections:

$$P(G) = 0.70(0.70) + 0.30\left(\frac{1}{6}\right) = 0.54,$$

$$P(B \mid G) = \frac{0.30(1/6)}{0.54} = \frac{0.05}{0.54} \approx 0.092593.$$

Exercise 10: Solution

For four inspections:

$$P(G) = 0.70(0.60) + 0.30\left(\frac{1}{14}\right) \approx 0.441429,$$

$$P(B \mid G) = \frac{0.30(1/14)}{0.70(0.60) + 0.30(1/14)} \approx 0.048544.$$

- Type B share among accepted lots: falls from about 9.26% to 4.85%
- Neither plan guarantees defect-free lots.

Exercise 10: Solution

- Inspection cost: every incoming lot
- Extra charge: only when the lot is both Type B and accepted

$$E(C_n) = 6n + 120P(B \cap G) = 6n + 120(0.30)P(G \mid B).$$

For three inspections:

$$E(C_3) = 18 + 120(0.30)\left(\frac{1}{6}\right) = 24.00 \text{ MAD}.$$

For four inspections:

$$E(C_4) = 24 + 120(0.30)\left(\frac{1}{14}\right) \approx 26.57 \text{ MAD}.$$

Measure	Inspect 3	Inspect 4
Acceptance probability	54.00%	44.14%
Type B probability among accepted lots	9.26%	4.85%
Expected cost per incoming lot	24.00 MAD	26.57 MAD

Exercise 10: Solution

- Recommend three inspections: this plan minimizes expected cost under the stated costs.
- Four inspections:
  - About 2.57 MAD more per incoming lot
  - Type B share of accepted lots lower by about 4.40 percentage points
  - More Type A lots rejected
- Choosing four instead would require an additional quality objective not specified in the question.